

FORMULA SHEET / FORMULEBLAD

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$CV = \left(\frac{s}{\bar{x}} \right) \cdot 100\%$$

If/as

$m_j = j^{\text{th}}$ midpoint / j^{de} middelpunt

$f_j = j^{\text{th}}$ frequency / j^{de} frekwensie

$c =$ Number of classes / Aantal klasse

then/dan

$$\bar{x} \approx \bar{x}_f = \frac{\sum_{j=1}^c m_j f_j}{n}$$

and/en

$$s^2 \approx s_f^2 = \frac{1}{n-1} \sum_{j=1}^c (m_j - \bar{x}_f)^2 f_j = \frac{1}{n-1} \left[\sum_{j=1}^c m_j^2 f_j - \frac{(\sum_{j=1}^c m_j f_j)^2}{n} \right]$$

$$\begin{aligned} s_{xy} &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n x_i y_i - \frac{(\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n} \right] \end{aligned}$$

$$r = \frac{s_{xy}}{s_x \cdot s_y}$$

$$P(A \text{ or/of } B) = P(A) + P(B) - P(A \text{ and/en } B)$$

$$P(A|B) = \frac{P(A \text{ and/en } B)}{P(B)}$$

Let A = simple event and $B_1, B_2, \dots, B_k$ = mutually exclusive and collectively exhaustive events. /

Laat A = eenvoudige gebeurtenis en $B_1, B_2, \dots, B_k$ = onderling uitsluitend en allesomvattende gebeurtenisse.

Then/Dan:

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_k)P(B_k)$$

and/en:

BAYES:

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_k)P(B_k)}$$

If X = discrete random variable:/As X = diskrete stogastiese veranderlike:

$$\mu = E[X] = \sum_{i=1}^N x_i P(x_i)$$

$$\sigma^2 = Var(X) = E[(X - \mu)^2] = \sum_{i=1}^N (x_i - \mu)^2 P(x_i) = \sum_{i=1}^N x_i^2 P(x_i) - \mu^2$$

If/As $X \sim \text{Binomial}(n, \pi)$:

$$P(X = x) = \frac{n!}{x!(n-x)!} \pi^x (1-\pi)^{n-x}; x = 0, 1, 2, \dots, n$$

$$\mu = n\pi$$

$$\sigma^2 = n\pi(1-\pi)$$

If/As $X \sim \text{Poisson}(\lambda)$:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, \dots$$

$$\mu = \sigma^2 = \lambda$$

If/As $X \sim \text{Hypergeometric}(n, N, E)$:

$$P(X = x) = \frac{\binom{E}{x} \binom{N-E}{n-x}}{\binom{N}{n}}; x = 0, 1, \dots, \min(E, n)$$

$$\mu = \frac{nE}{N}$$

$$\sigma^2 = \frac{nE(N-E)}{N^2} \cdot \frac{N-n}{N-1}$$

$$\begin{aligned} \text{Cov}(X, Y) &= \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] \\ &= \sum_{i=1}^N \sum_{j=1}^N (x_i - \mu_X)(y_j - \mu_Y) P(x_i, y_j) \\ &= \sum_{i=1}^N \sum_{j=1}^N x_i y_j P(x_i, y_j) - \mu_X \mu_Y \end{aligned}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

$$E[aX + bY] = aE[X] + bE[Y]$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

If/As $P = wX + (1 - w)Y$:

$$E(P) = wE(X) + (1 - w)E(Y)$$

$$\text{Var}(P) = w^2 \text{Var}(X) + (1 - w)^2 \text{Var}(Y) + 2w(1 - w) \text{Cov}(X, Y)$$

If/As $X \sim \text{Normal}(\mu, \sigma^2)$

$$E[X] = \mu; \text{Var}(X) = \sigma^2$$

$$Z = \frac{X - \mu}{\sigma} \sim \text{Normal}(0, 1)$$

If/As $X \sim \text{Uniform}(a, b)$

$$f(x) = \frac{1}{b-a}; a \leq x \leq b$$

$$E[X] = \frac{a+b}{2}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

If/As $X \sim \text{Exponential}(\lambda)$

$$f(x) = \lambda e^{-\lambda x}; x \geq 0$$

$$E[X] = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

$$P(X < x) = 1 - e^{-\lambda x}$$
